

Show understanding of the need for and purpose of ethics as a computing professional

9608_s21_qp_13_sc_7.c

QUESTION

(c) Patrick works for a company that has a code of conduct for its employees.

Explain the reasons why the company needs a professional code of conduct.

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MARK SCHEME

7(c)	1 mark per bullet point to max 3 e.g. • To identify what employees can do • To identify what employees cannot do • To identify the repercussions of employees performing activities they should not • To identify the company's/employee's responsibilities • To identify the company's values/missions • To identify what behaviour/events/activities can be measured against • To identify how they will comply with legislation	3
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9608_s20_qp_12_sc_4.a.i

QUESTION

4 Sophie is about to start a new job as a junior software developer.

(a) She is worried about joining a new team of people.

(i) State **one ethical** action that Sophie can take to help her to feel more confident about starting work.

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..... [1]

MARK SCHEME

4(a)(i)	1 mark only e.g. • Read about the languages she will be using • Visits the office prior to starting • Speaks to her manager about concerns	1
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9608_s20_qp_12_sc_4.a.ii

QUESTION

(ii) State **two ethical** actions that Sophie's manager can take to help Sophie to feel more confident about starting work.

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[2]

MARK SCHEME

4(a)(ii)	1 mark per action to max 2 For example: • Prepare an induction • Invite Sophie in before starting • Introduce Sophie to the team • Give Sophie a mentor	2
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9608_s20_qp_12_sc_4.b

QUESTION

(b) Explain why Sophie is asked to sign a professional code of conduct before starting work.

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[3]

MARK SCHEME

4(b)	1 mark per bullet point to max 3 - Sophie is confirming that she understands the code of conduct - To make sure Sophie knows what behaviour is expected of her - To make sure Sophie understands there may be consequences of some actions - To ensure all employees adhere to the same standards	3
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9608_s19_qp_13_sc_2.d

QUESTION

(d) One principle of the ACM/IEEE Software Engineering Code of Ethics is to always act in the best interest of the client.

Explain how Frankie can ensure that he is acting in the best interest of his client.

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MARK SCHEME

2(d)	1 mark per bullet point to max 3 For example: He shouldKeep the client's personal data private ...Involve the client in the development // ...Communicate with the client ...Provide the solution that the client asked for ...Keep the project running on time // budget ...Keep the client informed of any problems/delays	3
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QUESTION

4 A software developer works in a team for a large software development company.

(a) Two principles of the ACM/IEEE Software Engineering Code of Ethics are:

- developers must act consistently with the public interest
- developers must act in the best interest of their client and employer.

Name **and** describe **three** other principles in the ACM/IEEE Software Engineering Code of Ethics.

Principle 1

Description

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Principle 2

Description

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Principle 3

Description

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[6]

MARK SCHEME

4(a)	1 mark for naming a principle, 1 mark for description to max 3 · 2 Product Software engineers shall ensure that their products and related modifications meet the highest professional standards possible. Judgement Software engineers shall maintain integrity and independence in their professional judgement. Management Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance. Profession Software engineers shall advance the integrity and reputation of the profession consistent with the public interest. Colleagues Software engineers shall be fair to and supportive of their colleagues. Self Software engineers shall participate in life-long learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.	6
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9608_w17_qp_13_sc_6.a

QUESTION

- 6** Raj has joined a software company as a trainee programmer. He was given the company's Code of Conduct document during his induction training. The handbook has a section headed 'Ethical Behaviour'.

(a) Describe what is meant by **ethics**.

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[2]

MARK SCHEME

6(a)	Two marks from: A system of moral principles That guide behaviour / decision making Based on philosophical / religious views By example, e.g. respectful and considerate behaviour	Max 2
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QUESTION

(b) Raj is assigned to work as a new member of a development team.

In his first week, Raj feels uncomfortable working with one of his colleagues. He is unfamiliar with the programming language used by the team. Next week, he will be working on the site of one of the company's clients with a colleague. Raj is very nervous about working in an unfamiliar workplace.

Raj has a review with his manager after his first three weeks.

The Code of Conduct document was produced by the Human Resources section. It closely follows the ACM/IEEE Software Engineering Code of Ethics that uses these eight key principles:

Public	Client and Employer	Product	Judgement
Management	Profession	Colleagues	Self

There are issues Raj will want to raise with his manager.

- Describe **two** of these issues.
- Circle the key ACM/IEEE principle this comes under.
- Suggest what action should be taken to demonstrate ethical behaviour.

Issue 1

Description

.....

.....

ACM/IEEE principle (Circle one only)

Public	Client and Employer	Product	Judgement
Management	Profession	Colleagues	Self

Possible action

.....

.....

Issue 2

Description

.....

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ACM/IEEE principle (Circle one only)

Public

Client and Employer

Product

Judgement

Management

Profession

Colleagues

Self

Possible action

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[6]

MARK SCHEME

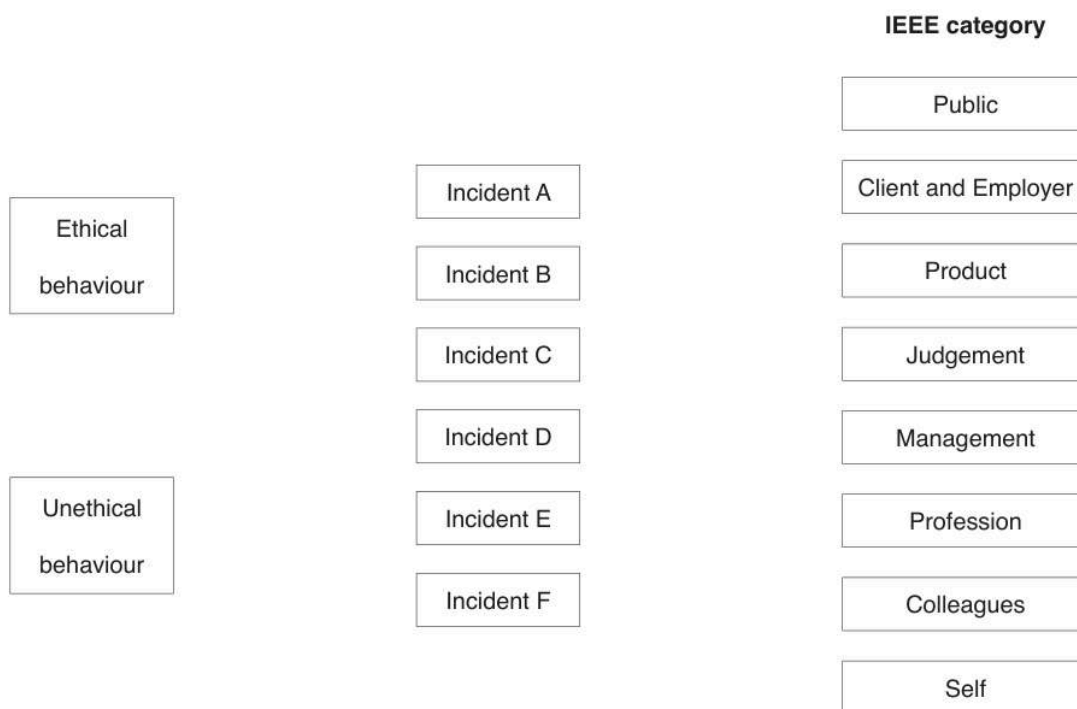
6(b)	One mark for identifying the issue One mark for correct principle One mark for possible action Max 2 issues (2 × 3 marks) 1 Uncomfortable with one of his colleagues Client and Employer // Management / Colleagues // Judgement // Self For example: Team building exercises // arranged meeting 2 Unfamiliar with programming language Self // Client and Employer //Product // Profession // Colleagues For example: Undergo training 3 Visit to unfamiliar workplace Client and employer // Management // Judgement // Profession // Colleagues For example: He should speak to his manager to discuss situation	Max 6
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QUESTION

- 5 The IEEE Software Engineering Code of Ethics uses eight key principles shown in the right-hand column of the following diagram.

Tom is employed as a tester with a software company. He is keen to become a trainee programmer.

The middle column in the diagram labels six incidents which have happened to Tom this week. The table that follows the diagram describes each incident.



Incident	Description
A	Tom has received some phishing emails. He reported this to the bank they were supposed to have come from.
B	Tom has asked his manager if they will pay for him to attend a programming course.
C	Tom is testing beta versions of new games software at work. He copies the software on to CD-Rs and sells them to his friends.
D	Tom has completed the application forms to join the Chartered Institute for IT.
E	Tom finds it difficult to work with one of his colleagues. His way of dealing with this has been to refuse to speak with the colleague.
F	Tom's manager had considered the testing of a new game was completed. Tom reported to his manager that he thought there were still bugs which needed to be rectified.

- (a) Draw a line on the diagram to link each of the six incidents to either ethical behaviour or unethical behaviour. [2]

MARK SCHEME

5(a)	Mark as follows: Unethical: C and E 1 Mark Ethical: A,B, D and F 1 Mark	2
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9608_w17_qp_12_sc_5.b

QUESTION

(b) Consider each incident you have identified as **ethical behaviour**.

Draw a line from each incident to indicate the IEEE category it maps to.

[4]

MARK SCHEME

5(b)	Mark as follows: A – Public interest B – Self D – Profession F – Product	1 Mark 1 Mark 1 Mark 1 Mark	4
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9608_s16_qp_12_sc_6

QUESTION

6 A team of software engineers is developing a new e-commerce program for a client.

State **three** of the principles of the ACM/IEEE Software Engineering Code of Ethics. Illustrate each one, with an example, describing how it will influence their working practices.

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.....[6]

MARK SCHEME

- 6** One mark for identifying the principle, **one mark** for an example that is in the context of this scenario.

Maximum of two marks per principle. **Maximum of three** principles.

[6]

- PUBLIC / Software engineers shall act consistently with the public interest.
 - Example in context
- CLIENT AND EMPLOYER / Software engineers shall act in a manner that is in the best interests of their client and employer (consistent with the public interest.)
 - Example in context
- PRODUCT / Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
 - Example in context
- JUDGEMENT / Software engineers shall maintain integrity and independence in their professional judgment.
 - Example in context
- MANAGEMENT / Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
 - Example in context
- PROFESSION / Software engineers shall advance the integrity and reputation of the profession (consistent with the public interest).
 - Example in context
- COLLEAGUES / Software engineers shall be fair to and supportive of their colleagues.
 - Example in context
- SELF / Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.
 - Example in context

QUESTION

- 6 A team of software engineers is developing a new e-commerce program for a client.

State **three** of the principles of the ACM/IEEE Software Engineering Code of Ethics. Illustrate each one, with an example, describing how it will influence their working practices.

1

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.....[6]

MARK SCHEME

- 6** **One mark** for identifying the principle, **one mark** for an example that is in the context of this scenario.

Maximum of two marks per principle. **Maximum of three** principles.

[6]

- PUBLIC / Software engineers shall act consistently with the public interest.
 - Example in context
- CLIENT AND EMPLOYER / Software engineers shall act in a manner that is in the best interests of their client and employer (consistent with the public interest.)
 - Example in context
- PRODUCT / Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
 - Example in context
- JUDGEMENT / Software engineers shall maintain integrity and independence in their professional judgment.
 - Example in context
- MANAGEMENT / Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
 - Example in context
- PROFESSION / Software engineers shall advance the integrity and reputation of the profession (consistent with the public interest).
 - Example in context
- COLLEAGUES / Software engineers shall be fair to and supportive of their colleagues.
 - Example in context
- SELF / Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.
 - Example in context

Show understanding of the need to act ethically and the impact of acting ethically or unethically for a given situation

9608_s21_qp_13_sc_7.c

QUESTION

(c) Patrick works for a company that has a code of conduct for its employees.

Explain the reasons why the company needs a professional code of conduct.

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..... [3]

MARK SCHEME

7(c)	1 mark per bullet point to max 3 e.g. • To identify what employees can do • To identify what employees cannot do • To identify the repercussions of employees performing activities they should not • To identify the company's/employee's responsibilities • To identify the company's values/missions • To identify what behaviour/events/activities can be measured against • To identify how they will comply with legislation	3
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QUESTION

6 This question presents three scenarios.

Tick (✓) **one** box for each scenario to indicate whether you think the named person's behaviour is ethical or unethical. Justify your choice.

- (a) Latifah has changed jobs and has started to work for a new company. The company uses an Integrated Development Environment (IDE) to develop code. Latifah decides not to use the IDE that the company has because she is familiar with a different IDE.

Ethical	
Unethical	

Justification

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..... [2]

MARK SCHEME

6(a)	1 mark per bullet point for each justification, to max 2 e.g. Either Ethical - Latifah believes that her productivity is better using an IDE that she has experience of ... and the work is still compatible with other IDEs ...the IDE is open source so does not cost anything extra - Reference to IEEE standards <u>in context</u> Or Unethical - Latifah is knowingly using software that is not provided or supported by the company ...and this may lead to issues with future software compatibility/maintenance - Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (b) Samid is in charge of a project to write a banking application. He is employing staff to work on the application. His daughter is a computer security expert. She is looking for a new job. Samid decides to employ his daughter.

Ethical	
Unethical	

Justification

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..... [2]

MARK SCHEME

6(b)	1 mark per bullet point for each justification, to max 2 e.g. Either Ethical • Samid believes his daughter is qualified and would be the best person for the job • ...and believes that it is in the best interest of his client to secure a high-quality employee as soon as possible • ... they may have interviewed her with a third party and she came out as the best candidate • Reference to IEEE standards <u>in context</u> Or Unethical • Samid shows poor judgement by favouring his daughter • ...rather than using a fair and open recruitment process • there may be people with better qualifications that are being overlooked • Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (c) Jason works for a social media company. He is concerned that users of the company's social media website have not been told how their personal data is being used.

Jason tells his manager his concerns. His manager tells him not to worry because there have been no complaints from the users. Jason takes no further action.

Ethical	
Unethical	

Justification

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MARK SCHEME

6(c)	1 mark per bullet point for each justification, to max 2 e.g. Either Ethical • Jason has already raised his concerns (to his manager) • ...and has fulfilled his professional responsibilities by doing so. • Reference to IEEE standards <u>in context</u> Or Unethical • Jason should continue to raise any concerns he has with a higher level of management in the company • ...because it may be against the public interest for users to have their data used in this way without them explicitly knowing it. • Reference to IEEE standards <u>in context</u>	2
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QUESTION

7 This question presents three scenarios.

Tick (✓) **one** box for each scenario to indicate whether you think the behaviour shown is ethical or unethical. Justify your choice.

- (a) Marina has a very short deadline to create a piece of software for a client. Algorithm A is quick to code and simpler to test but has an inefficient run time. Algorithm B is more complex to code and test but has a more efficient run time. She decides to implement algorithm A.

Ethical	
Unethical	

Justification

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MARK SCHEME

7(a)	1 mark per bullet point for each justification, to max 2 Either Unethical • Marina knowingly produces a solution that is not as efficient as it could be • ...and this does not enhance the reputation of the person/company • Reference to IEEE standards <u>in context</u> Or Ethical • Marina acts in the best interests of the employer to complete a working product on time • ...that is more likely to be correct as the coding is simpler • Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (b) Doug is managing a project for a client that has fallen behind schedule. He asks all members of the project team to work extra hours and to cancel any holiday plans to get the project back on schedule.

Ethical	
Unethical	

Justification

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MARK SCHEME

7(b)	1 mark per bullet point for each justification, to max 2 Either Unethical • Doug has a management responsibility • ...to be fair to his colleagues • ...but is showing a lack of personal integrity by applying pressure • Reference to IEEE standards <u>in context</u> Or Ethical • Doug is trying to act in the best interests of his employer/client • ...and may be asking employees to voluntarily offer to work additional hours • ...he may be paying them extra • Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (c) Debbie is programming a car safety management system. She thinks that the test plan she has been given is not adequate. She decides to discuss her concerns on a public internet forum.

Ethical	
Unethical	

Justification

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MARK SCHEME

7(c)	1 mark per bullet point for each justification, to max 2 Either Unethical • Debbie may be breaking company policy discussing current projects // confidentiality • ...and has a duty to act in the best interests of his company • ...and should have raised her concerns internally • Reference to IEEE standards <u>in context</u> Or Ethical • Debbie may feel that in the public interest the problem should be discussed • ...to ensure that a safety critical system is fully tested • ...so as to prevent potential loss of life • Reference to IEEE standards <u>in context</u>	2
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QUESTION

- 4** Sophie is about to start a new job as a junior software developer.

- (a) She is worried about joining a new team of people.

- (i) State **one ethical** action that Sophie can take to help her to feel more confident about starting work.

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..... [1]

MARK SCHEME

4(a)(i)	1 mark only e.g. • Read about the languages she will be using • Visits the office prior to starting • Speaks to her manager about concerns	1
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9608_s20_qp_12_sc_4.a.ii

QUESTION

(ii) State **two ethical** actions that Sophie's manager can take to help Sophie to feel more confident about starting work.

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[2]

MARK SCHEME

4(a)(ii)	1 mark per action to max 2 For example: • Prepare an induction • Invite Sophie in before starting • Introduce Sophie to the team • Give Sophie a mentor	2
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9608_s20_qp_12_sc_4.a.iii

QUESTION

(iii) State **one ethical** action that Sophie's new colleagues can take to help Sophie to feel more confident about starting work.

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MARK SCHEME

4(a)(iii)	1 mark only e.g. • Prepare a greeting/introduction • Provide structured support • Invite Sophie to social event(s) before/at the start to meet people	1
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9608_s20_qp_12_sc_4.b

QUESTION

(b) Explain why Sophie is asked to sign a professional code of conduct before starting work.

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MARK SCHEME

4(b)	1 mark per bullet point to max 3 • Sophie is confirming that she understands the code of conduct • To make sure Sophie knows what behaviour is expected of her • To make sure Sophie understands there may be consequences of some actions • To ensure all employees adhere to the same standards	3
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QUESTION

- (d) One principle of the ACM/IEEE Software Engineering Code of Ethics is to always act in the best interest of the client.

Explain how Frankie can ensure that he is acting in the best interest of his client.

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..... [3]

MARK SCHEME

2(d)	1 mark per bullet point to max 3 For example: He shouldKeep the client's personal data private ...Involve the client in the development // ...Communicate with the client ...Provide the solution that the client asked for ...Keep the project running on time // budget ...Keep the client informed of any problems/delays	3
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QUESTION

4 A software developer works in a team for a large software development company.

(a) Two principles of the ACM/IEEE Software Engineering Code of Ethics are:

- developers must act consistently with the public interest
- developers must act in the best interest of their client and employer.

Name **and** describe **three** other principles in the ACM/IEEE Software Engineering Code of Ethics.

Principle 1

Description

.....

.....

Principle 2

Description

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Principle 3

Description

.....

.....

[6]

MARK SCHEME

4(a)	1 mark for naming a principle, 1 mark for description to max 3 · 2 Product Software engineers shall ensure that their products and related modifications meet the highest professional standards possible. Judgement Software engineers shall maintain integrity and independence in their professional judgement. Management Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance. Profession Software engineers shall advance the integrity and reputation of the profession consistent with the public interest. Colleagues Software engineers shall be fair to and supportive of their colleagues. Self Software engineers shall participate in life-long learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.	6
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9608_w18_qp_13_sc_3.a

QUESTION

- 3** This question presents three scenarios. Tick (✓) **one** box for each scenario to indicate whether you think the person's behaviour is ethical or unethical. Justify your choice.

- (a)** Mason is using his work computer to book a holiday whilst at work.

Ethical	☐
Unethical	☐

Justification

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.....[2]

MARK SCHEME

3(a)	Either Ethical • He is booking the holiday in his own time / lunchtime // he is self-employed • He has been given permission • Reference to IEEE <u>in context</u> Or Unethical • Should not use company computer for personal use • Should be working whilst at work • Reference to IEEE <u>in context</u>	2
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9608_w18_qp_13_sc_3.b

QUESTION

(b) Ethan is supervising a trainee. The trainee asks Ethan for a reference for another job. Ethan does not want to lose the trainee, so refuses to give him a reference.

Ethical	
Unethical	

Justification

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.....[2]

MARK SCHEME

3(b)	Unethical: Max 2 marks from • Company will get a bad reputation • Should be supporting his colleague • Reference to IEEE <u>in context</u>	2
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QUESTION

- (c) Margarita finds that one of her team members has produced some inventive code. She presents this to her manager, praising the individual by name.

Ethical	
Unethical	

Justification

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.....[2]

MARK SCHEME

3(c)	Either Ethical • She is supporting her colleague • Working in the best interests of the company • Reference to IEEE <u>in context</u> Or Unethical • Praising one team member instead of the whole team • Others in the team may also have contributed, so she is not being supportive of the whole team • Reference to IEEE <u>in context</u>	2
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QUESTION

- 5 This question presents three scenarios. For each scenario, tick (✓) **one** box to show whether you think the person's behaviour is ethical or unethical. Justify your choice.

- (a) Wendy is a software engineer who is developing a program for her company. Her friend, Noah, is developing a program for a different company. Wendy looks at the code that Noah is writing to get ideas for her own program.

Ethical	☐
Unethical	☐

Justification

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..... [2]

MARK SCHEME

5(a)	1 mark per bullet point for each justification, to max 2 Either Unethical - Noah's work may be confidential - Wendy shouldn't claim someone else's ideas / work as her own - She is bringing the profession into disrepute - Reference to IEEE standards <u>in context</u> Or Ethical - The code could be open source - Wendy may have permission from Noah - Wendy isn't copying the code, just getting ideas - Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (b) Amit is fixing some bugs in the computer system of a large multinational company. He is asked to sign a confidentiality agreement. He sees some confidential information which contains the names of other multinational companies that have broken the law. He copies this information and releases it on the Internet.

Ethical	
Unethical	

Justification

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..... [2]

MARK SCHEME

5(b)	1 mark per bullet point for each justification, to max 2 Either Unethical • Amit has a responsibility to his company • He should have taken it to the police rather than putting it on the Internet • He has a signed agreement to say he will not give anything away • Reference to IEEE standards <u>in context</u> Or Ethical • Amit is acting in the public interest • Amit may not have actually signed the confidentiality agreement • If acting illegally, the multinational company should be brought to justice • Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (c) Farah is providing a company with an estimate for the cost of writing a program. The company she works for is in financial difficulty so she increases the estimate by 10%.

Ethical	
Unethical	

Justification

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..... [2]

MARK SCHEME

5(c)	1 mark per bullet point for each justification, to max 2 Either Ethical - It might save people's jobs - Farah is acting in the best interest of her company - Reference to IEEE standards <u>in context</u> Or Unethical - Farah has a responsibility to act in the best interest of her client - It could give her company a bad reputation - Reference to IEEE standards <u>in context</u>	2
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9608_w18_qp_11_sc_5.a

QUESTION

5 For each of the following scenarios, tick (✓) **one** box for each scenario to indicate whether you think the person's behaviour is ethical or unethical. Justify your choice.

(a) Kevin is a software engineer who has recently started a job with a new company. He is using program code from his previous employer in his new employer's programs.

Ethical	☐
Unethical	☐

Justification

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.....[2]

MARK SCHEME

5(a)	1 mark per bullet point for each justification, to max 2 Either Unethical - Work belongs to the company it was created for // copyright ... Kevin cannot use it without permission - It reduces the integrity of the person / profession / new company - Reference to IEEE standards <u>in context</u> Or Ethical - The program code could be open source - Kevin might own the copyright of code - Kevin may have permission to use the code - Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (b) Nadya is a software developer. She has accepted a new job. She has never worked with the programming languages used by this new company. Nadya is planning to increase her knowledge of these programming languages before she starts her new job.

Ethical	
Unethical	

Justification

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.....[2]

MARK SCHEME

5(b)	1 mark per bullet point for each justification, to max 2 Either Unethical Nadya has accepted a role / work she knows she cannot do This reduces the integrity of the person She may let down the new organisation who are expecting her to be able to do the work Reference to IEEE standards <u>in context</u> Or Ethical She is taking steps to prepare for the role ... Without expecting the company to do it Nadya may have told the company that she didn't know the languages but that she would learn them Reference to IEEE standards <u>in context</u>	2
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QUESTION

- (c) Maria finds that one of her team members has produced some inventive code. She presents this to her manager, stating that it was produced by the team. She does not mention the individual's name.

Ethical	
Unethical	

Justification

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.....[2]

MARK SCHEME

5(c)	1 mark per bullet point for each justification, to max 2 Either Ethical - The individual works as part of the team therefore, the team should / will get the credit - Maria is not lying about who produced it - Reference to IEEE standards <u>in context</u> Or Unethical - Maria should identify who / where the idea originated - It prevents the individual getting recognition - Maria is not being supportive of the individual - Reference to IEEE standards <u>in context</u>	2
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QUESTION

(b) Raj is assigned to work as a new member of a development team.

In his first week, Raj feels uncomfortable working with one of his colleagues. He is unfamiliar with the programming language used by the team. Next week, he will be working on the site of one of the company's clients with a colleague. Raj is very nervous about working in an unfamiliar workplace.

Raj has a review with his manager after his first three weeks.

The Code of Conduct document was produced by the Human Resources section. It closely follows the ACM/IEEE Software Engineering Code of Ethics that uses these eight key principles:

Public	Client and Employer	Product	Judgement
Management	Profession	Colleagues	Self

There are issues Raj will want to raise with his manager.

- Describe **two** of these issues.
- Circle the key ACM/IEEE principle this comes under.
- Suggest what action should be taken to demonstrate ethical behaviour.

Issue 1

Description

.....

.....

ACM/IEEE principle (Circle one only)

Public	Client and Employer	Product	Judgement
Management	Profession	Colleagues	Self

Possible action

.....

.....

Issue 2

Description

.....

.....

ACM/IEEE principle (Circle one only)

Public

Client and Employer

Product

Judgement

Management

Profession

Colleagues

Self

Possible action

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[6]

MARK SCHEME

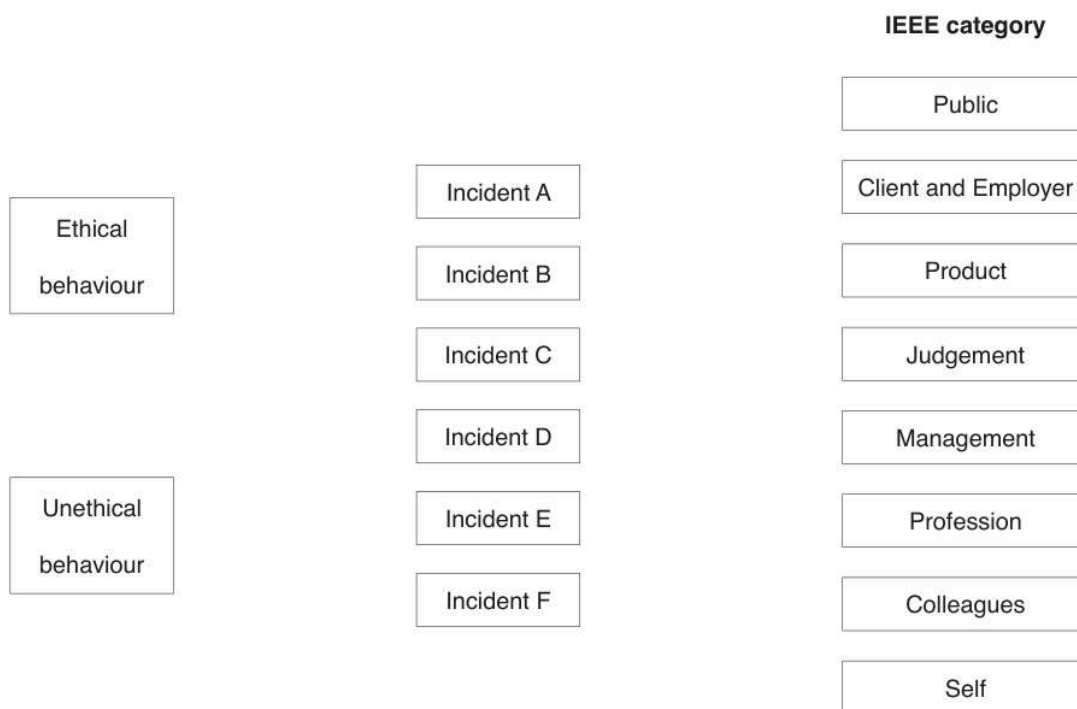
6(b)	One mark for identifying the issue One mark for correct principle One mark for possible action Max 2 issues (2 × 3 marks) 1 Uncomfortable with one of his colleagues Client and Employer // Management / Colleagues // Judgement // Self For example: Team building exercises // arranged meeting 2 Unfamiliar with programming language Self // Client and Employer //Product // Profession // Colleagues For example: Undergo training 3 Visit to unfamiliar workplace Client and employer // Management // Judgement // Profession // Colleagues For example: He should speak to his manager to discuss situation	Max 6
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QUESTION

- 5 The IEEE Software Engineering Code of Ethics uses eight key principles shown in the right-hand column of the following diagram.

Tom is employed as a tester with a software company. He is keen to become a trainee programmer.

The middle column in the diagram labels six incidents which have happened to Tom this week. The table that follows the diagram describes each incident.



Incident	Description
A	Tom has received some phishing emails. He reported this to the bank they were supposed to have come from.
B	Tom has asked his manager if they will pay for him to attend a programming course.
C	Tom is testing beta versions of new games software at work. He copies the software on to CD-Rs and sells them to his friends.
D	Tom has completed the application forms to join the Chartered Institute for IT.
E	Tom finds it difficult to work with one of his colleagues. His way of dealing with this has been to refuse to speak with the colleague.
F	Tom's manager had considered the testing of a new game was completed. Tom reported to his manager that he thought there were still bugs which needed to be rectified.

- (a) Draw a line on the diagram to link each of the six incidents to either ethical behaviour or unethical behaviour. [2]

MARK SCHEME

5(a)	Mark as follows: Unethical: C and E 1 Mark Ethical: A,B, D and F 1 Mark	2
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9608_w17_qp_12_sc_5.b

QUESTION

(b) Consider each incident you have identified as **ethical behaviour**.

Draw a line from each incident to indicate the IEEE category it maps to.

[4]

MARK SCHEME

5(b)	Mark as follows: A – Public interest B – Self D – Profession F – Product	1 Mark 1 Mark 1 Mark 1 Mark	4
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9608_s16_qp_12_sc_6

QUESTION

6 A team of software engineers is developing a new e-commerce program for a client.

State **three** of the principles of the ACM/IEEE Software Engineering Code of Ethics. Illustrate each one, with an example, describing how it will influence their working practices.

1

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2

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3

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.....[6]

MARK SCHEME

- 6** One mark for identifying the principle, **one mark** for an example that is in the context of this scenario.

Maximum of two marks per principle. **Maximum of three** principles.

[6]

- PUBLIC / Software engineers shall act consistently with the public interest.
 - Example in context
- CLIENT AND EMPLOYER / Software engineers shall act in a manner that is in the best interests of their client and employer (consistent with the public interest.)
 - Example in context
- PRODUCT / Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
 - Example in context
- JUDGEMENT / Software engineers shall maintain integrity and independence in their professional judgment.
 - Example in context
- MANAGEMENT / Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
 - Example in context
- PROFESSION / Software engineers shall advance the integrity and reputation of the profession (consistent with the public interest).
 - Example in context
- COLLEAGUES / Software engineers shall be fair to and supportive of their colleagues.
 - Example in context
- SELF / Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.
 - Example in context

QUESTION

- 6 A team of software engineers is developing a new e-commerce program for a client.

State **three** of the principles of the ACM/IEEE Software Engineering Code of Ethics. Illustrate each one, with an example, describing how it will influence their working practices.

1

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2

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3

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.....[6]

MARK SCHEME

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 - Example in context
- SELF / Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.
 - Example in context

QUESTION

- 4 Paul works part-time for a large software company. The company sells security software to a number of banks. He also runs his own software company that produces and sells computer games.

Six statements about computer ethics are shown below.

Draw lines to indicate whether each statement describes ethical or unethical behaviour.

Statement

To save time, Paul fakes the test results when testing the bank security software.

Paul uses the software developed in his day job to help write some of the games software routines.

To allow him to concentrate on his games software, Paul has frequently turned down job opportunities in his day job.

To make the games software more realistic, Paul uses password protection code used in the bank security software.

Because his work load is increasing, Paul is now using overseas companies to write some of the routines used in his games software.

Paul carries out training on how to write games software in his spare time.

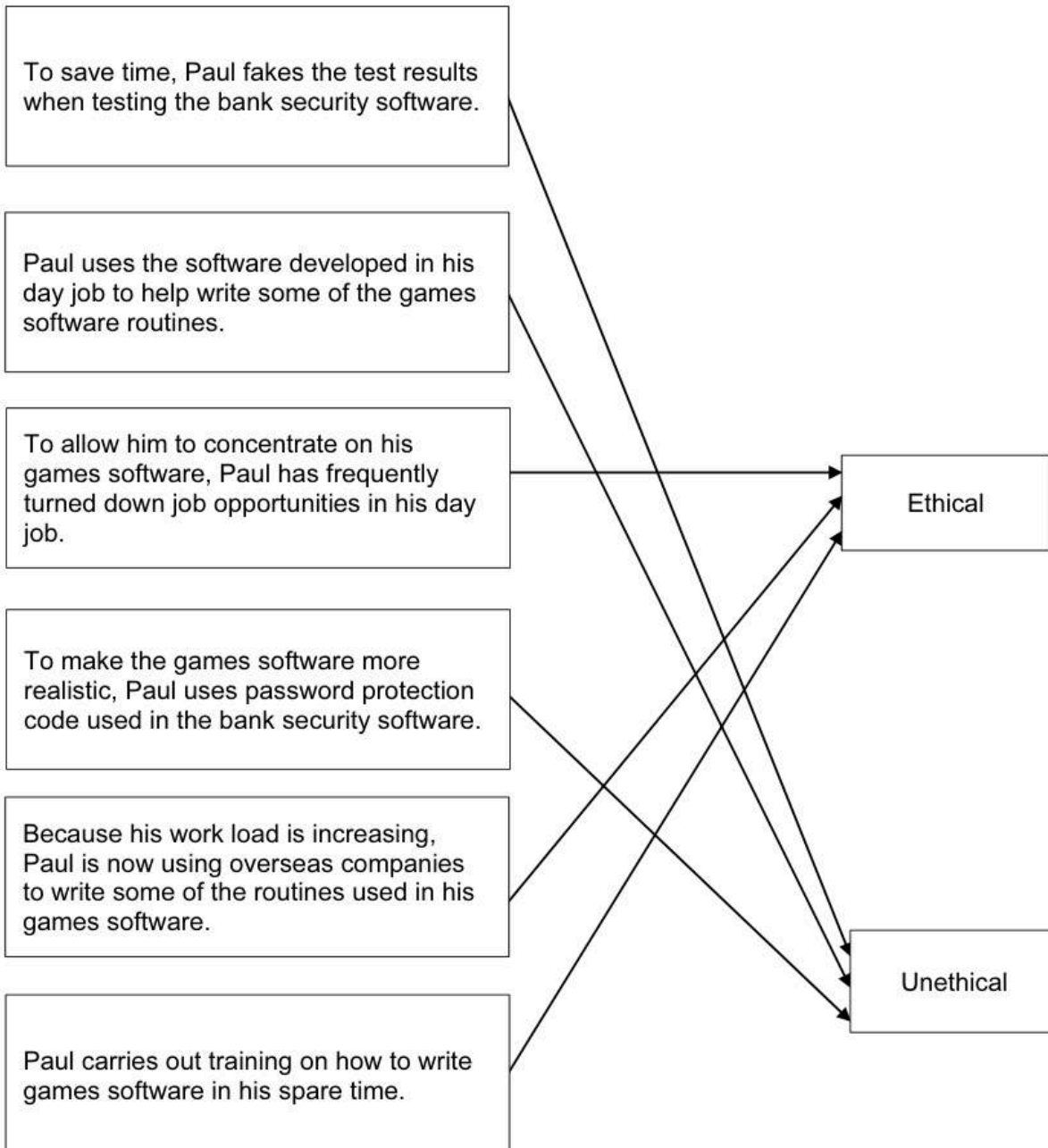
Ethical

Unethical

[6]

MARK SCHEME

4



1 mark for each correct line, two lines from one box is incorrect

[6]

QUESTION

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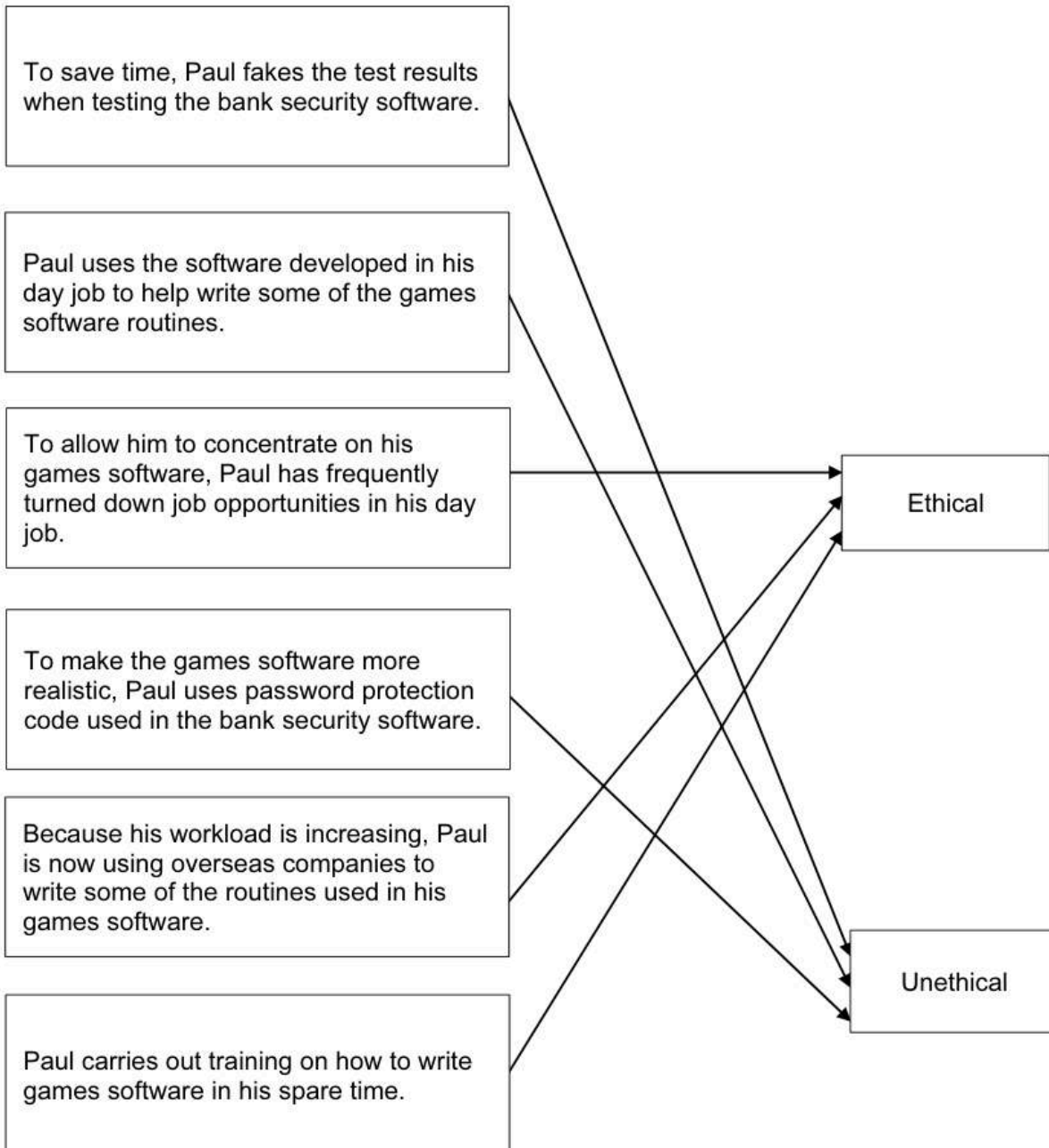
Ethical

Unethical

[6]

MARK SCHEME

4



1 mark for each correct line, two lines from one box is incorrect

[6]

Show understanding of the need for copyright legislation

9608_w19_qp_12_sc_5.a

QUESTION

5 Mica has created some software and has copyrighted it. She wants to stop other people from copying and changing it illegally.

(a) Identify **two** ways Mica can prevent illegal copies of the software being installed.

1

.....

2

.....

[2]

MARK SCHEME

5(a)	1 mark per bullet point to max 2 Encryption Assign a unique code so it will not install without this // product key Limit the number of times that version of the software can be installed Set a time limit within which the software must be installed	2
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9608_w19_qp_12_sc_5.b

QUESTION

(b) Identify **one** way Mica can distribute the software without the source code.

.....

..... [1]

MARK SCHEME

5(b)	Provide an .exe file // Compile the source code // Use a compiler	1
------	---	----------

QUESTION

(d) Shazia wants to make sure her computer program is copyrighted.

(i) Define the term **copyright**.

.....
..... [1]

MARK SCHEME

4(d)(i)	1 mark for correct answer Formal or legal recognition of ownership of the program // Formal or legal restriction / permissions on use of the program // The intellectual property rights to the program	1
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Show understanding of the different types of software licencing and justify the use of a licence for a given situation

9608_s21_qp_13_sc_7.b

QUESTION

- (b) Patrick has completed the application. He needs to choose whether to distribute the software application using an open source licence or a commercial licence.

Describe **open source** and **commercial** software licensing.

Open source

.....

.....

.....

Commercial

.....

.....

.....

[4]

MARK SCHEME

7(b)	1 mark per bullet point, max 2 marks per licence Open Source • The (source) <u>code</u> is distributed with the software • users can modify the software • The software is (usually) free of cost • Users must redistribute the modified software under the same terms Commercial • The software is (usually) purchased for a cost • A licence defines how it can be used • The source code is not distributed // users cannot change the software	4
------	---	----------

QUESTION

(c) Annchi needs to decide which type of software licence to use for the game.

(i) Give **two** benefits to **Annchi** of using a commercial licence.

1

.....

2

.....

[2]

MARK SCHEME

4(c)(i)	1 mark per benefit to max 2 • She can charge a fee for the game • She retains the copyright • ... so, the game cannot be re-distributed by a third-party without her permission	2
---------	--	----------

QUESTION

(ii) Give **one** benefit to the **customers** of the game being released using a commercial licence.

.....

..... [1]

MARK SCHEME

4(c)(ii)	1 mark per benefit to max 1 • Potentially better support, as she is charging a fee • Likely to have fewer bugs / less prone to malware than if distributed under other licences e.g. open source • Redress available if the game does not function correctly	1
----------	---	----------

QUESTION

- (iii) Describe **one** benefit to the **customers** of the game being released using a shareware licence.

.....

.....

.....

..... [2]

MARK SCHEME

4(c)(iii)	1 mark per bullet point • They can check it works // check if it meets their requirements • ... without having to paying a fee if it does not	2
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QUESTION

- (c) Mica is releasing the software under a commercial licence.

- (i) Give **two** benefits to Mica of using a commercial licence.

1

.....

2

..... [2]

MARK SCHEME

5(c)(i)	1 mark per benefit to max 2 So that she can sell the software for a fee // make money from the software A commercial licence prohibits unauthorised/further copies being made and/or distributed A commercial licence prohibits any changes to the software	2
---------	--	----------

QUESTION

(ii) Name **two** other types of software licence.

1

.....

2

.....

[2]

MARK SCHEME

5(c)(ii)	1 mark per bullet point to max 2 Open Source Free Software Shareware Freeware	2
----------	---	----------

QUESTION

(ii) Shazia does not want to release the software as open source.

Explain why Shazia does **not** want to use an open source licence.

.....

.....

.....

..... [2]

MARK SCHEME

4(d)(ii)	1 mark per bullet point • She does not wish to release the source <u>code</u> • She does not want anyone to be able to edit / modify / share the source code/program • She wants to make money from the program	2
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QUESTION

- (iii) Name **and** describe **two** software licences, other than open source that Shazia could use.

Licence 1

.....

.....

Licence 2

.....

.....

[2]

MARK SCHEME

4(d)(iii)	1 mark for a <u>name and a description</u> of each licence to max 2 Commercial Software • The program is purchased for a fee • It restricts the number of users/possible time period for use // Limited number of installations allowed // Software key needed to install • Source code not provided // source code protected / cannot be edited • Anyone can purchase/download if agree to the terms Shareware • The program is free for a trial period // The (free) program may have limited functionality // Need to purchase / enter details after trial • Users do not have access to the source code // source code may not be edited • Users may re-distribute the software. Freeware • There is no charge for the software • The software could still be copyrighted • She can set her own restrictions on what a user can do with the program	2
-----------	--	----------

QUESTION

- 2 A software company produces software and distributes it under different software licences.

(a) Four descriptions of software licences are given.

Write the type of software licence that best fits each description. Use a different type of licence for each description.

1. The software can be legally used, only after a fee has been paid.

Licence type

2. The source code comes with the software. If the software is modified, the edited source code must be released under the same conditions as the original software.

Licence type

3. The software is free for a trial period and then a fee is requested, or expected, if the user wants to continue to use the software.

Licence type

4. The source code comes with the software. The software is free to be downloaded, edited, and distributed, possibly without restriction.

Licence type

[4]

MARK SCHEME

2(a)	1 mark for each correct term Commercial Licence Free Software Licence Shareware Licence Open Source Licence	4
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QUESTION

- 5 Arnold is a software developer. He has created a computer game for people to download over the Internet. Arnold is considering releasing the game as a piece of commercial software.

(a) (i) Describe what is meant by a **commercial licence**.

.....

 [2]

MARK SCHEME

5(a)(i)	1 mark per bullet point, max 2 • Restricted use • Source code not provided // source code protected • Anyone can purchase/download if agree to the terms • Limited number of installations allowed // Software key needed to install	2
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9608_w18_qp_13_sc_5.a.ii

QUESTION

(ii) Name and describe **one** other type of licence that Arnold can consider using.

Licence type

Description

.....

.....

.....

[3]

MARK SCHEME

5(a)(ii)	1 mark for name and 2 marks for description Either Open Source • The source code is released with the program • Users can change / edit the source code to enhance the game • Users can re-release the game under the same terms // the game might spread more easily Or Shareware • Users get free trial or limited access for set time • Users do not have access to the source code // source code may not be edited • At end of trial period, users may have to pay or register to continue using the game // Can get people 'hooked' and then charge a fee Or Freeware • There is no fee for the game • The game could be copyrighted • Modification, re-distribution or reverse engineering of the game without permission is prohibited	3
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QUESTION

- 3 Hugo has produced a program (app) for mobile phones. He needs to decide whether to use an Open Source licence or to distribute the app as shareware.

(a) Describe what is meant by **Open Source licence** and **shareware**.

Open Source

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.....

.....

Shareware

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.....

.....

[4]

MARK SCHEME

3(a)	1 mark per bullet point, max 2 marks per licence Open Source - The source code is released with program - Users can edit the source code to suit their needs - Users re-release their version under the same terms - Can be cost-free but may also need payment Shareware - Users get a free trial of the software ... which may be limited in features - No access to source code // Program cannot be edited - Then they have to pay / sign-up after the expiry date // Then they have to pay / sign-up to get full functionality // Then they have to pay / sign-up to stop unwanted pop-ups, etc.	4
------	--	----------

QUESTION

(b) Tick (✓) **one** box to indicate the licence Hugo should use. Justify your choice.

Open Source	☐
Shareware	☐

Justification

.....

.....

.....

.....[2]

MARK SCHEME

3(b)	1 mark per bullet point to max 2 marks for chosen licence Open Source For example: Hugo does not have to set up ways to take funds Others may enhance / improve the program Hugo can charge a fee for the App <u>Or</u> Shareware For example: Hugo can charge for the App Not giving away the code/people can't copy the code ... Hugo gets the sole recognition for the program Possible legal consequences if someone does copy the code If users need to sign up, their data can be used for marketing etc. Customers have peace of mind that the software hasn't been maliciously edited / bugs introduced	2
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9608_s16_qp_13_sc_3.a

QUESTION

- 3** A company needs new software to manage its accounts. It is evaluating two different options. One option is open source software and the other is commercial software.

(a) Explain what is meant by open source software.

.....

.....

.....

.....[2]

MARK SCHEME

3 (a) Two from: [2]

- The source code comes with the software.
- The user can edit the source code.
- Once edited, the software is re-distributed with the changes.

9608_s16_qp_13_sc_3.b

QUESTION

(b) Explain what is meant by commercial software.

.....

.....

.....

.....[2]

MARK SCHEME

(b) Two from: [2]

- The software is purchased.
- With a **licence** which restricts the number of users / possible time period for use.
- The program code for the software cannot be edited.

QUESTION

(c) The company has decided to purchase commercial software.

Identify **four** benefits to the company in choosing the commercial software option.

- 1
.....
- 2
.....
- 3
.....
- 4
.....[4]

MARK SCHEME

(c) **Four** from:

[4]

- Support / training is readily available so help can be accessed if needed.
- More robust software / fewer bugs as it has been tested more thoroughly/by more users.
- Forums / user groups will exist for popular software.
- Software upgrade path likely to be available (at minimal cost).
- Manufacturer develops patches that can be automatically downloaded.
- Compatibility is inbuilt for other commercial software.

QUESTION

- 5 Three types of software licensing and four descriptions are shown in the table below.

Put a tick (✓) in each row to match each description to the appropriate type of software licensing.

Description	Type of software		
	Open source	Shareware	Commercial
Software is purchased before it can be used			
Source code comes with the software			
Software is provided free on a trial basis			
The software can be modified by the user			

[4]

MARK SCHEME

- 5 One mark per row.

[4]

No mark if more than one tick in any row.

Description	Open source	Shareware	Commercial
Software is purchased before it can be used			✓
Source code comes with the software	✓		
Software is provided free on a trial basis		✓	
The software can be modified by the user	✓		

QUESTION

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The software can be modified by the user			

[4]

MARK SCHEME

- 5 One mark per row.

[4]

No mark if more than one tick in any row.

Description	Open source	Shareware	Commercial
Software is purchased before it can be used			✓
Source code comes with the software	✓		
Software is provided free on a trial basis		✓	
The software can be modified by the user	✓		

Show understanding of Artificial Intelligence (AI)
